

# Final Model

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## Final Model

After spending weeks searching for the best model to forecast Initial Claims in SC, I believe the best model will be using data from 2001 (excludes S&P500) and predict  $\log(\text{InitialClaims Per Capita})$  with the predictors: Unemployment Per Capita, Labor Force Per Capita, and Layoffs.

## Data

```
# data is from the CleaningData.rmd file that combines all datasets used into one final dataset
data <- read.csv("Datasets/FinalData.csv", header=T)

# convert date to date format
data$Date <- as.Date(data$Date)

# subset of data starting in 2001
data.2001 <- data %>%
  filter(Date >= "2001-01-01",
         !year(Date) %in% c(2020, 2025),
         Date < "2024-12-31")

# check data
tail(data.2001, 5)
```

| ##     | Date                        | Population         | InitialClaims                | InitialClaims.PerCapita |
|--------|-----------------------------|--------------------|------------------------------|-------------------------|
| ## 271 | 2024-07-31                  | 5478831            | 13851                        | 0.002528094             |
| ## 272 | 2024-08-31                  | 5478831            | 9458                         | 0.001726281             |
| ## 273 | 2024-09-30                  | 5478831            | 7520                         | 0.001372556             |
| ## 274 | 2024-10-31                  | 5478831            | 11596                        | 0.002116510             |
| ## 275 | 2024-11-30                  | 5478831            | 7785                         | 0.001420924             |
| ##     | InitialClaims.PerChange     | InitialClaims.Lag  | InitialClaims.Lag.PerCapita  |                         |
| ## 271 | 33.43931                    | 9458               | 0.001726281                  |                         |
| ## 272 | -31.71612                   | 7520               | 0.001372556                  |                         |
| ## 273 | -20.49059                   | 11596              | 0.002116510                  |                         |
| ## 274 | 54.20213                    | 7785               | 0.001420924                  |                         |
| ## 275 | -32.86478                   | 13464              | 0.002457459                  |                         |
| ##     | InitialClaims.Lag.PerChange | InitialClaims.Lag2 | InitialClaims.Lag2.PerCapita |                         |
| ## 271 | -31.71612                   | 7520               | 0.001372556                  |                         |
| ## 272 | -20.49059                   | 11596              | 0.002116510                  |                         |
| ## 273 | 54.20213                    | 7785               | 0.001420924                  |                         |

```
## 274          -32.86478          13464          0.002457459
## 275          72.94798          14667          NA
##      InitialClaims.Lag2.PerChange Unemployment Unemployment.Per
## 271          -20.490590          97764          8.968100
## 272          54.202128          107891          10.358619
## 273          -32.864781          113975          5.639025
## 274          72.947977          118097          3.616583
## 275          8.934938          121482          2.866288
##      Unemployment.PerCapita Employment Employment.Per Employment.PerCapita
## 271          0.01784395          2415158          0.20741370          0.4408163
## 272          0.01969234          2414624          -0.02211035          0.4407188
## 273          0.02080280          2413300          -0.05483255          0.4404772
## 274          0.02155515          2413165          -0.00559400          0.4404525
## 275          0.02217298          2412589          -0.02386907          0.4403474
##      LaborForce LaborForce.Per LaborForce.PerCapita UnemploymentRate
## 271          2512922          0.5218257          0.4586603          3.890451
## 272          2522515          0.3817468          0.4604112          4.277120
## 273          2527275          0.1887006          0.4612800          4.509798
## 274          2531262          0.1577589          0.4620077          4.665538
## 275          2534071          0.1109723          0.4625204          4.793946
##      UnemploymentRate.Per SP500 SP500.Per SP500_Health SP500_Health.Per
## 271          8.402428 5522.30 1.1321349          1742.67          2.4901049
## 272          9.938930 5648.40 2.2834688          1829.71          4.9946347
## 273          5.440059 5762.48 2.0196870          1796.48          -1.8161348
## 274          3.453376 5705.45 -0.9896781          1711.50          -4.7303616
## 275          2.752261 6032.38 5.7301352          1713.71          0.1291265
##      SP500_Energy SP500_Energy.Per LFPR LFPR.Per Hires Hires.Per Separations
## 271          712.42          2.0322816 62.7 0.0000000          5230 -4.0543020          5170
## 272          695.86          -2.3244715 62.7 0.0000000          5519 5.5258126          5197
## 273          676.47          -2.7864800 62.5 -0.3189793          5350 -3.0621489          5285
## 274          681.25          0.7066093 62.5 0.0000000          5307 -0.8037383          5087
## 275          724.06          6.2840367 62.5 0.0000000          5374 1.2624835          5082
##      Separations.Per Quits Quits.Per Layoffs Layoffs.Per Openings Openings.Per
## 271          -4.85829960          3177 -6.283186          1697 -2.750716          7649          1.932303
## 272          0.52224371          3085 -2.895814          1844          8.662345          7103          -7.138188
## 273          1.69328459          3217 4.278768          1788          -3.036876          7615          7.208222
## 274          -3.74645222          3032 -5.750699          1739          -2.740492          8031          5.462902
## 275          -0.09828976          3095 2.077836          1669          -4.025302          7508          -6.512265
```

## Model Creation

```
# model
final.model <- lm(log(InitialClaims.Lag.PerCapita) ~ Unemployment.PerCapita + LaborForce.PerCapita + Layoffs.PerCapita, data = data.2001)

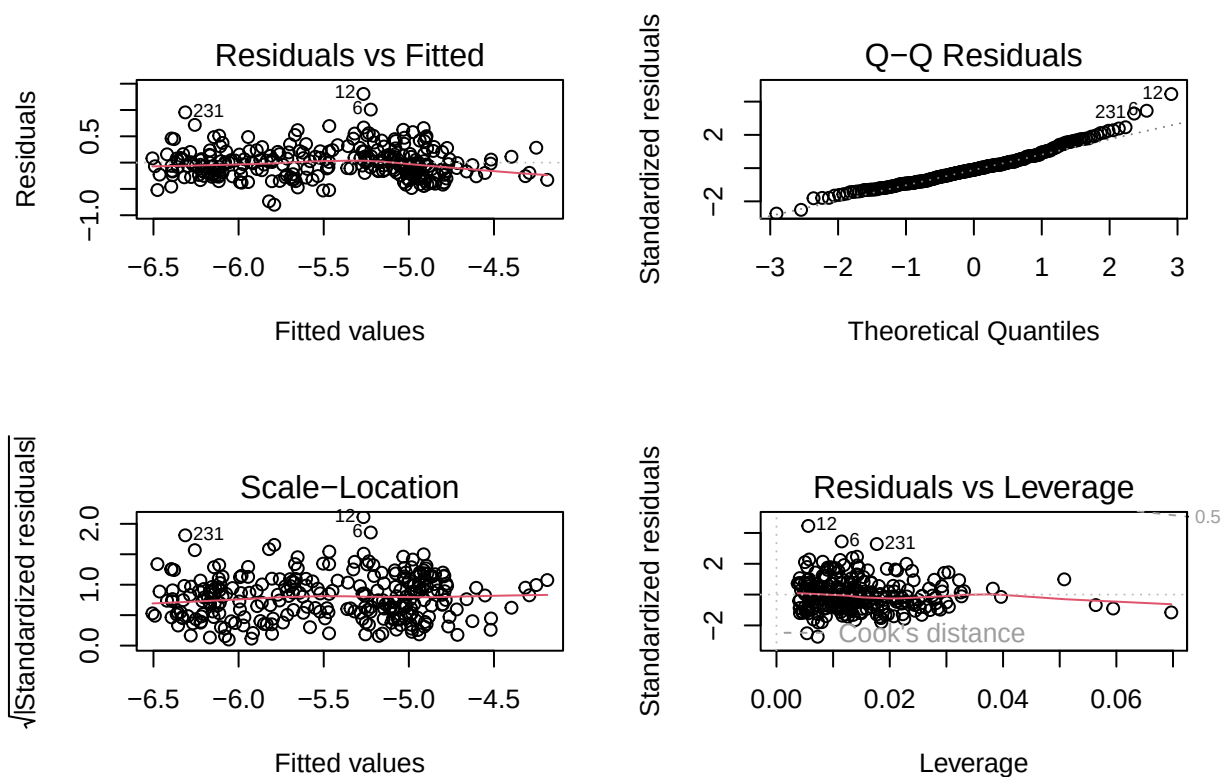
# check model
summary(final.model)

##
## Call:
## lm(formula = log(InitialClaims.Lag.PerCapita) ~ Unemployment.PerCapita +
##      LaborForce.PerCapita + Layoffs, data = data.2001)
##
```

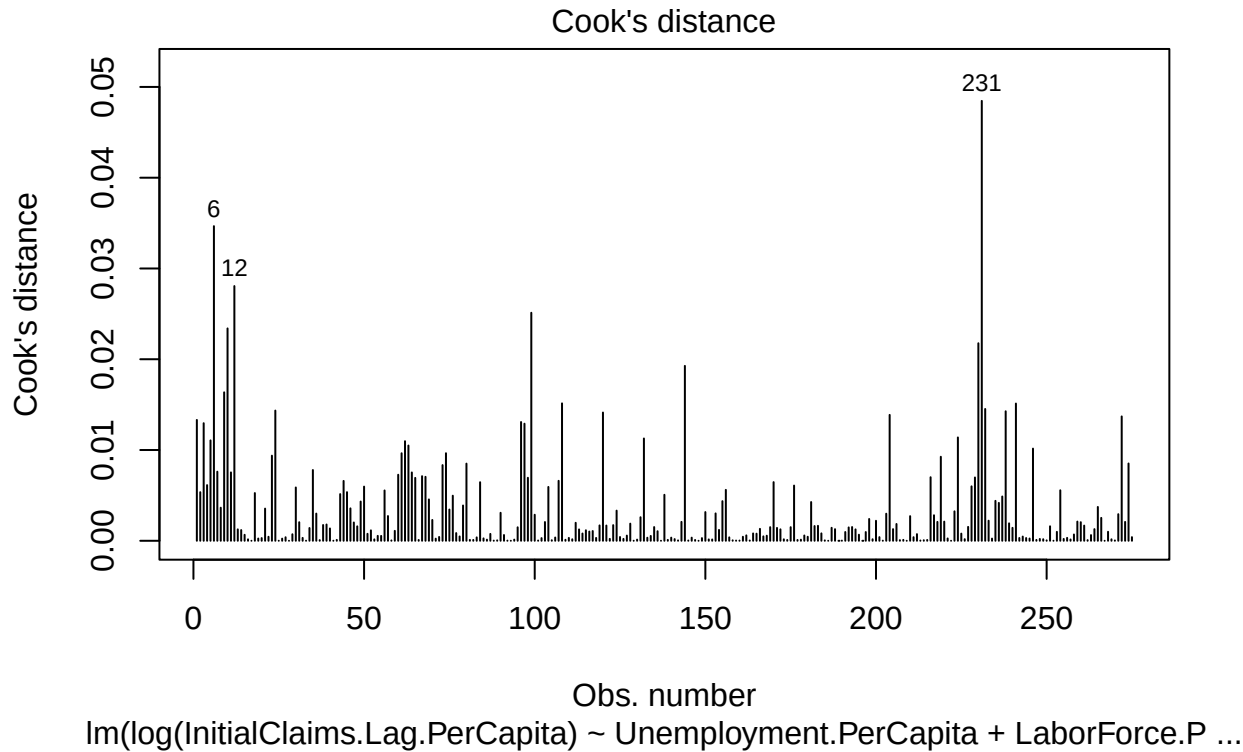
```
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.80062 -0.21361 -0.02854  0.15028  1.30515
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -1.656e+01  7.402e-01 -22.365  < 2e-16 ***
## Unemployment.PerCapita  1.858e+01  1.835e+00  10.125  < 2e-16 ***
## LaborForce.PerCapita   1.894e+01  1.754e+00  10.801  < 2e-16 ***
## Layoffs           9.093e-04  1.059e-04   8.585  7.24e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2938 on 271 degrees of freedom
## Multiple R-squared:  0.7865, Adjusted R-squared:  0.7841
## F-statistic: 332.7 on 3 and 271 DF,  p-value: < 2.2e-16
```

## Diagnostic Plots

```
# plot diagnostic plots
par(mfrow=c(2,2))
plot(final.model)
```



```
# check for outliers
par(mfrow=c(1,1))
plot(final.model, 4)
```



Check Multicollinearity

```
# check vif values
print(vif(final.model))
```

|                           |                      |          |
|---------------------------|----------------------|----------|
| ## Unemployment.PerCapita | LaborForce.PerCapita | Layoffs  |
| ## 1.577356               | 1.692791             | 1.639066 |

Check Confidence Intervals

```
# confidence intervals
confint(final.model)
```

|                           |               |               |
|---------------------------|---------------|---------------|
| ##                        | 2.5 %         | 97.5 %        |
| ## (Intercept)            | -1.801279e+01 | -15.098112092 |
| ## Unemployment.PerCapita | 1.496873e+01  | 22.194631540  |
| ## LaborForce.PerCapita   | 1.549142e+01  | 22.397758105  |
| ## Layoffs                | 7.007725e-04  | 0.001117853   |

## Check Model's Effectiveness at Predicting Initial Claims

### Splitting into Training and Testing Data

```
# set seed
set.seed(112233)

# split the data (70-30)
train.indices <- createDataPartition(data.2001$InitialClaims.Lag.PerCapita, p = 0.7, list = FALSE)

train.data <- data.2001[train.indices, ]
test.data <- data.2001[-train.indices, ]

# create training model
training.model <- lm(log(InitialClaims.Lag.PerCapita) ~ Unemployment.PerCapita + LaborForce.PerCapita +

# create predictions
training.model.pred <- predict(training.model, newdata = test.data)

# MSE and RMSE
MSE <- mean((test.data$InitialClaims.Lag.PerCapita - training.model.pred)^2)
RMSE <- sqrt(MSE)

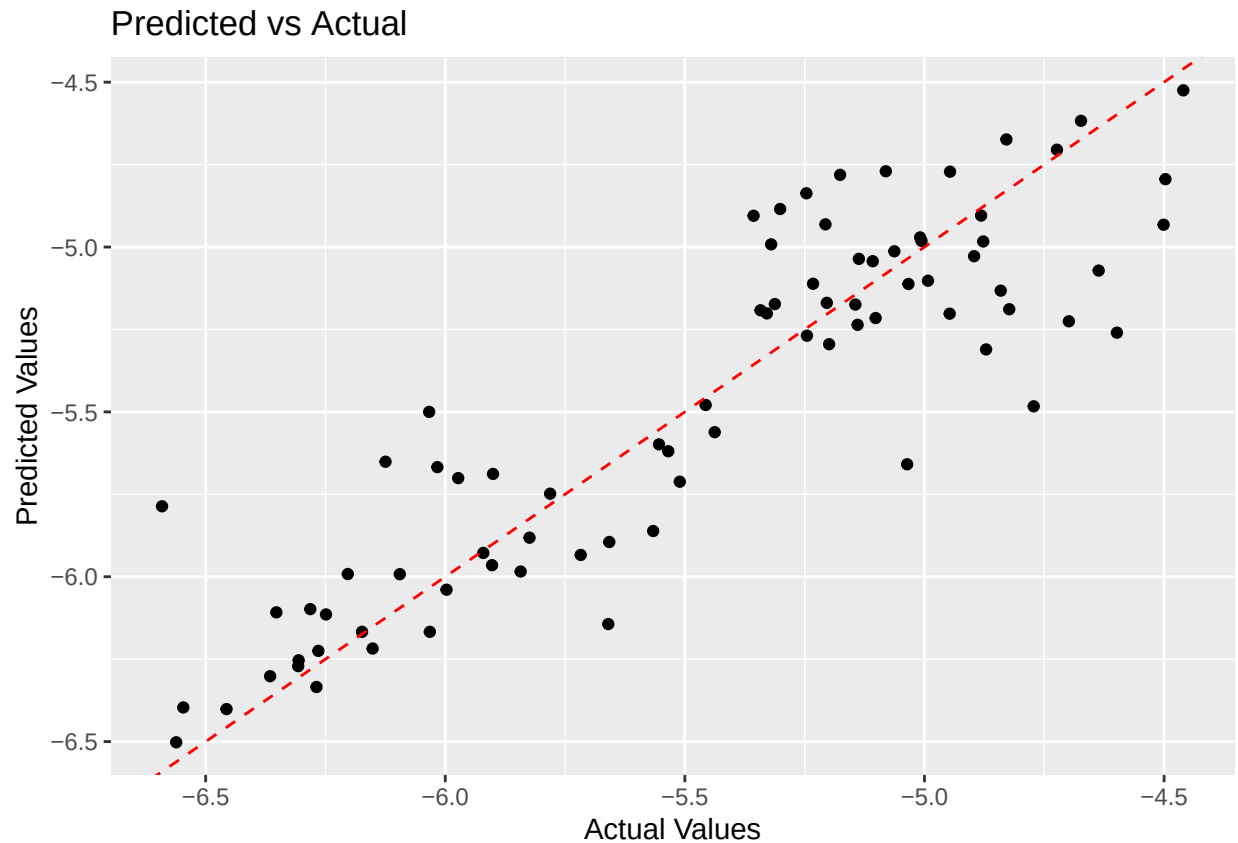
print(MSE)
```

```
## [1] 30.26826
```

```
print(RMSE)
```

```
## [1] 5.50166
```

```
# plot predictions
ggplot(test.data, aes(x = log(InitialClaims.Lag.PerCapita), y = training.model.pred)) +
  geom_point() +
  geom_abline(intercept = 0, slope = 1, color = "red", linetype = "dashed") +
  labs(title = "Predicted vs Actual",
       x = "Actual Values",
       y = "Predicted Values")
```



### Plotting Predicted Initial Claims over Actual Initial Claims

```
# gather predictions
predictions <- predict(final.model, newdata = data.2001)

# fix corrections
predictions <- exp(predictions)

# make new variable in dataset for the predicted intial claims
data.2001$PredictedInitialClaimsLaggedPerCapita <- predictions

# Un-Lag the Predictions
data.2001$PredictedInitialClaimsPerCapita <- lead(data.2001$PredictedInitialClaimsLaggedPerCapita, 1)

# Un-Per Capita the Predictions
data.2001$PredictedInitialClaims <- data.2001$PredictedInitialClaimsPerCapita * data.2001$Population

# check data
# print(data.2001, 5)

# create plot with both actual and predicted initial claims
ggplot(data.2001, aes(x = Date)) +
  # add recession bars
  geom_rect(aes(xmin=as.Date("2001-3-01"), xmax=as.Date("2001-11-01"), ymin=-Inf, ymax=Inf), fill="red2")
```

```

geom_rect(aes(xmin=as.Date("2007-12-01"), xmax=as.Date("2009-06-01"), ymin=-Inf, ymax=Inf), fill="red",
# add rest
geom_line(aes(y = InitialClaims, color = "Actual Initial Claims")) +
geom_line(aes(y = PredictedInitialClaims, color = "Predicted Initial Claims")) +
labs(title = "Actual vs Predicted Initial Claims", color = "color",
      x = "Date",
      y = "Initial Claims") +
scale_color_manual(values = c("Actual Initial Claims" = "steelblue4", "Predicted Initial Claims" = "brown"))
theme_fivethirtyeight()

```

## Warning: Removed 1 row containing missing values or values outside the scale range  
 ## ('geom\_line()').

